Biomedical Engineering Laboratory I (Fall 2025)

Lab 4 Biomaterials and Drug Delivery[1]

OBJECTIVES

1. Fabricate and characterize a hydrogel-based delivery system.
2. Measure kinetics of dye release from the hydrogel-delivery system.

BACKGROUND

Biomaterials and Biocompatibility

A biomaterial is a living or non-living material that interacts with a biological environment without compromising the inherent biological or physiological functionality of the environment. A biomaterial is used to diagnose, treat, support, augment or replace part or all of a living system. While in addition to demonstrating efficacy for its intended function(s), a biomaterial must satisfy a plethora of requirements to be successfully integrated with native tissues, many of the parameters can be described by one word: biocompatibility. Biocompatibility is the ability of a biomaterial to perform with an appropriate host response in a specific application, without activating chronic inflammation pathways or an immune response. Host response characterizes the behavior of the biological environment to the implanted material.

Hydrogels as Biomaterials

Hydrogels are polymeric materials that actively interact with a hydrated environment. A hydrogel is made up of two phases; a solid biopolymer and an aqueous phase comprised largely of water. Examples of hydrogel materials include alginate, chitosan, and synthetic polymers such as polyethylene glycol. Alginate is a biopolymer isolated from brown algae or seaweed. It is found commercially as a thickener in many food products such as ice cream and salad dressing. Alginate is synthesized by crosslinking its monomers, D-mannuronic acid and L-guluronic acid (see Figure 1). The alginate

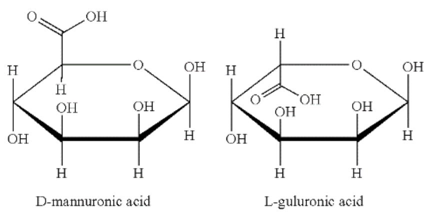

Figure 1. Monomer structures of the alginate molecule.

[1] Adapted from Helen H. Lu, Columbia University.

polymer chains are cross-linked by divalent ions such as calcium and magnesium. The crosslinking process can be employed in two ways: external and internal gelation. The external gelation method describes the diffusion of divalent ions into the alginate matrix; Alginate solution being extruded as droplets into a solution with divalent ions results in the formation of gel alginate beads. Inversely, divalent ions solution being extruded into alginate solution results in gel capsules. For internal gelation, an insoluble calcium salt is added to the alginate-drug solution and the mixture extruded into oil. The latter is acidified to bring about the release of Ca^{2+} from the insoluble salt for cross-linking with the alginate.

Alginate is a well-established biomaterial. It is biocompatible, non-immunogenic and biodegradable. Alginate-based cell encapsulation systems and drug delivery systems have been widely investigated for biomedical applications. Alginate has been used to form a 3-D culturing environment for cells like chondrocytes, which are responsible for the formation and maintenance of articular cartilage. In tissue engineering, these systems have also been considered for encapsulation of pancreatic cells and liver cells, providing a steady, biological source of factors important in maintaining functionality, induction of tissue repair and regeneration. Encapsulation of cells provides protection from immune cells and unwanted release of cell products that are modulated by external stimuli (e.g., insulin release by pancreatic beta cells in response to variations in blood glucose levels).

Alginate beads and capsules have also been shown to be beneficial as drug delivery systems, e.g., for the delivery of antibodies, the nicotine patch, and insulin delivery systems in pancreatic islets. The alginate functions as a targeted delivery system, which is advantageous for a variety reasons including:

1. The alginate systems are designed to act locally, minimizing systemic side effects associated with oral or intravascular drug delivery;
2. They are cost effective as less of the drug is needed as it is administered locally;
3. Alginate beads or capsules can be manufactured at room temperature, thus the bioactivity of drugs/growth factors and cells are not compromised;
4. The crosslinking process of alginate can be systematically controlled and optimized to produce beads and capsules of various diameters, wall thicknesses, levels of permeability, and stability.

Alginate systems are not without disadvantages: the low stability and high porosity of the bead or capsule membrane limits the application of alginate in extended delivery and load bearing applications. Despite this limitation, these systems have been researched extensively for the treatment of infections, cancer, and diabetes.

In this lab, you will fabricate an alginate-based delivery system. The alginate will be loaded with a model dye, which you will use to quantify the diffusive release from each alginate type, emulating how a drug would be delivered from an alginate system.

MATERIALS AND METHODS

- Sodium alginate solution (1% and 2%)
- 4% $CaCl_2$ solution
- Deionized water
- Dextran
- Trypan blue
- Microscope
- Microplate reader
- Magnetic stirrer and stir plate
- Ring stand with clamp
- 1 mL syringes
- Needles (#12, #6, and #4 $\frac{1}{2}$)
- Graduated cylinder
- Centrifuge tubes (5 mL and 50 mL)
- Petri dishes (60 mm/100 mm)
- 24-well and 96-well plate(s)
- Pipettes
- Rulers, Calipers
- Timers

LABORATORY PROCEDURES

For this experiment, you will use various formulations of alginate and $CaCl_2$ solutions. Since we will be examining diffusion of trypan blue as a model for drug delivery, you will need to first load alginate beads and capsules with the dye. In these preparatory steps, you will create the solutions necessary to perform the subsequent experiments. **Upon the completion of preparatory procedures 1-4, you will have: (a) created a standard curve for trypan blue solution, (b) created the solutions for experiments, (c) fabricated and characterized alginate beads and capsules, and (d) designed an experiment to characterize dye release from alginate beads and capsules.**

Procedure 1: Construct a Standard Curve

Using a microplate reader, you will obtain quantitative values that correspond to the absorbance of solutions. In order to convert absorbance values into actual concentrations, a standard curve must be derived. The standard curve entails measuring the absorbance (y-axis) of known concentrations (x-axis), then applying a linear fit to this data. Here are the basic steps that you will use to create a standard curve.

1. Make at least five (5) solutions of trypan blue in deionized water. The concentrations should range from 0% to 5% (v/v) of the stock solution of trypan blue.
2. Transfer the dilutions of trypan blue (100 ～ 200 μL) to a 96-well plate with the deionized water as the blank; each concentration should occupy a separate well(n=3).
3. Using the microplate reader, measure the absorbance of each solution against the blank. The absorbance at 630 nm will be measured.
4. From the microplate reader data, plot absorbance vs. concentration.
5. Apply a linear fit to the absorbance vs. concentration data. The resultant curve can be used to determine concentration from absorbance data in subsequent analysis.

Note that the standard curve obtained here may not necessarily be applicable to your experimental data; the dynamic range of your absorbance could be significantly less than this curve. Therefore, it may be necessary to generate another curve that has enough resolution at concentrations comparable to the color changes observed in your experiment. Keep this in mind when designing your experimental protocol for Procedure 5.

Procedure 2: Solution Preparation

1. In a 5 mL tube, make a 4 mL solution of 10% trypan blue (v/v) mixed with 1% Alginate (the alginate solution has been prepared and will be provided to you by TA.)
2. In a 5 mL tube, make a 4 mL solution of 10% trypan blue (v/v) mixed with 2% Alginate (the alginate solution has been prepared and will be provided to you by TA.)
3. In a 5 mL tube, make a 4 mL solution of 10% trypan blue mixed with 4% $CaCl_2$ ($CaCl_2$ solution has been prepared and will be provided to you by TA.)
4. (optional)In a 5 mL tube, make a 4 mL solution of 10% trypan blue, 5% Dextran (w/v, g/mL), mixed with 4% $CaCl_2$ solution.

Procedure 3: Alginate Bead Fabrication

1. Fill a petri dish more than halfway with 4% $CaCl_2$ solution.
2. Draw 1 mL of alginate solution + trypan blue into a 1 mL syringe and remove all air.

3. Attach a syringe with a #4$\frac{1}{2}$(26 G) needle and hold the syringe over the $CaCl_2$ surface at a measured height.
4. Slowly push out the alginate solution until a drop of alginate falls into the $CaCl_2$ solution. Move the syringe slightly over the $CaCl_2$ surface and repeat until at least 10 beads have been formed (Figure 2). **Make sure the bead is fully immersed in solution**.

Figure 2. Trypan blue-loaded alginate beads.

5. Allow the beads to sit in the solution for ~ 5 minutes and then measure the beads' diameters. It is up to you to determine which tools and methods will be used to determine bead diameter. Hint: A caliper is probably not the best choice as it will deform the bead or capsule.
6. Attach a #6(23 G) needle to the syringe and position the syringe over a new dish of the $CaCl_2$ solution. Create at least 10 beads and measure their diameters.
7. Repeat protocol with a #12(18 G) needle. *What is the effect of needle size on bead diameter?*
8. Be sure to document all of the settings of your system so that you can reproduce beads of approximately the same size in the following procedures. *Do the variations in bead sizes appear to be significantly different? What is the bead size distribution? Use the appropriate statistical testing to support your answer*.
9. Use the microscope to observe beads structure, *save pictures when necessary*.

Procedure 4: Alginate Capsule Fabrication

1. Place the alginate solution and a small magnetic stir bar in a beaker.
2. Turn on the stirrer to a moderate speed.
3. Mix the $CaCl_2$/trypan mixture thoroughly and draw 1 mL of $CaCl_2$/trypan mixture into a syringe.
4. Place a #4$\frac{1}{2}$ (26 G) needle on the syringe and remove all air from the needle tip.
5. Hold the syringe with the tip of the needle above the alginate solution. Find the optimal distance from the alginate and the vortex at which the syringe should be placed and record measurement.
6. Very slowly push out the $CaCl_2$ solution until a drop falls out.
7. Make several more capsules with the same dropping condition (Figure 3) and allow it to spin for a few minutes (a little longer than that for beads).

8. Change with different type of needles and dropping variables until spherical capsules are obtained. Turn off stir plate and transfer capsules to a container of deionized water labeled with drop speed, stir speed, drop height, drop distance from vortex, syringe angle, and time in the corresponding alginate solution.

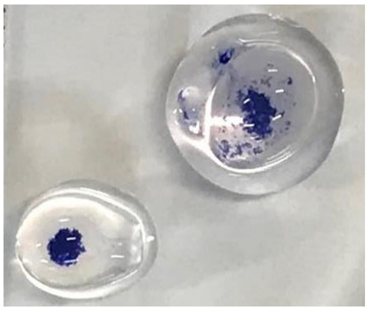

Figure 3. Trypan blue-loaded alginate capsule.

9. (optional) Repeat the procedure, replacing the trypan + $CaCl_2$ solution with the trypan + $CaCl_2$ + Dextran solution. Adding dextran increases the viscosity of the solution and allows it to penetrate the surface tension of the alginate during dropping. <u>*How do you think that this will affect the shape and stability of the capsules?*</u>
10. Use the microscope to observe capsule structure, <u>*save pictures when necessary.*</u> Measure the capsule diameter and wall thicknesses. <u>*How can you vary capsule wall thickness?*</u>

Procedure 5: Measure Dye Release from Alginate Beads and/or Capsules

During this procedure, you will utilize the parameters derived in the previous procedures to rapidly create enough beads or capsules to perform an experiment of your own design. Before embarking on your experiment, re-check the quality of fabricated beads/capsules and make adjustments as necessary. Upon completion of Procedure 5, you will have: (a) Fabricated and characterized alginate beads of two different sizes (or two degree of crosslinking) and capsules of two different wall thicknesses, and (b) Measured dye release from alginate beads and/or capsules.

Using the settings that you determined in Procedures 1-4, you are to devise an experiment that can be used to quantify the kinetics of dye release from alginate beads and capsules. The independent variables in this experiment, i.e., the parameters of interest are bead/capsule size, the degree of crosslinking, and time in solution. You are to devise an experimental protocol that you will implement to quantify trypan blue dye release from beads of varying sizes, different degree of crosslinking, beads vs. capsules, or effects of capsule wall thickness on dye release. You will fabricate a number of beads or capsules and place them in 24-well plates or beakers with deionized water. At selected time points, you will transfer solution from the beads/capsules to an empty well of a 96-well plate for microplate reader. The diffusion will be quantified by the color of the transferred solution using a microplate reader (for details, see microplate reader tutorial on Blackboard). In creating your protocol, be sure to include the following sections:

- Experimental Objective(s) – major question(s) that you are striving to address and their biomedical significance.

- Hypothesis – the theory/expectation that you will be testing via the experiment
- Experimental Design
  - Experimental variables – what parameters will be adjusted for testing the hypothesis
  - Experimental and control groups with the number of replications or samples per group
  - Experimental methodology & outcome parameters measured – step by step methodology that will be conducted and what will be measured.
  - Method of analyses – what techniques and devices will be used to measure outcomes
  - Statistics analysis – statistical techniques that will be used to draw conclusions from the data

After performing your experiment, consider the following questions in your report:

1. *What is the primary mechanism driving the dye release?*
2. *Which factors (carrier- and release media-related) determine the kinetics of dye release?*
3. *How does the release kinetics differ between alginate beads and capsules? Statistically compare the release for beads and capsules of comparable size and shape.*
4. *Do you observe a zero order or first order release?*
5. *What else must be considered if using growth factors instead of the model dye?*

Laboratory Report Guidelines

- Be sure to provide sufficient details of your protocols for quantifying dye release in the Materials and Methods section in your report.
- Provide sample calculation of experiment actual concentration from standard curve.
- Discuss the significance of alginate beads and capsules with regards to biomaterials, drug delivery, etc.
- Be sure to answer all the questions listed in the procedure.

References/Relevant Reading

1. Blandino *et. al.*; Glucose oxidase release from calcium alginate gel capsules. *Enzyme Microb. Tech.*, **2000**, 27: 319-324.
2. Kikuchi *et. al.*; Effect of Ca^{2+}-alginate gel dissolution on release of dextran with different molecular weights. *J. Control Release*, **1999**, 58: 21-28.

3. Lu *et. al*.; Controlled delivery of platelet-rich plasma-derived growth factors for bone formation. *J. Biomed. Mater. Res.,* **2008***,* 86A: 1128–1136.